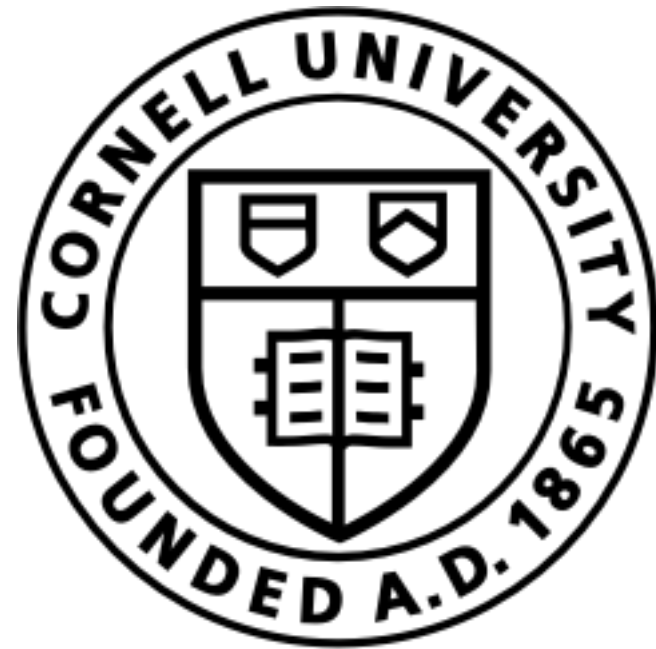




Lecture 3b: Numerical Solution of Optimal Rational Choice Problems and the Role of Prices, Budgets and Information

CHEME 5760 Cornell University

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Objectives. In this lecture, we will:

- ▶ **O1:** Introduce Constrained Resource Allocation Problems. Constrained resource allocation problems are found in a huge variety of disciplines, ranging from the sciences and engineering to hospitality.
- ▶ **O2:** The role of budget (and potentially) other constraints in resource allocation problems. Constraints shrink the space of possibilities to a specific feasible region.
- ▶ **O3:** Introduce the Lagrange multiplier method for solving constrained resource allocation problems: Maximize Utility subject to a budget constraint
- ▶ L3a and L3b long-form notes: <https://bit.ly/45EInfO>

Jupyter Notebooks

Repository: <https://github.com/varnerlab/CHEME-5760-Examples-F23.git>

- Example: CHEME-5760-L3b-OptimalSoln-CobbDouglas-Utility.ipynb
- Example: CHEME-5760-OptimalSoln-Linear-Utility.ipynb
- Example: CHEME-5760-L3b-OptimalSoln-KompalaModel.ipynb

Concept: Constrained Resource Allocation Problems

An agent has a set of n objects $X = \{x_i\}_{i=1}^n$, a utility function $U : X \rightarrow \mathbb{R}$, and a total of I units of resource to allocate, e.g., money, time, etc. An optimal rational agent maximizes its utility subject to its resource budget:

$$\text{maximize } \mathcal{O} = U(x_1, \dots, x_n) \quad (1)$$

$$\text{subject to } \sum_{i \in 1, \dots, n} c_i x_i = I \quad (2)$$

$$\text{and } x_i \geq 0 \quad i = 1, 2, \dots, n \quad (3)$$

The quantity $c_i \geq 0 \forall i$ denotes the cost of object i , and $x_i \geq 0$ represents the amount of object i purchased or consumed by the agent.

Concept: The Grocery Store Fruit Problem



$$c_1 = 0.98$$

Apples



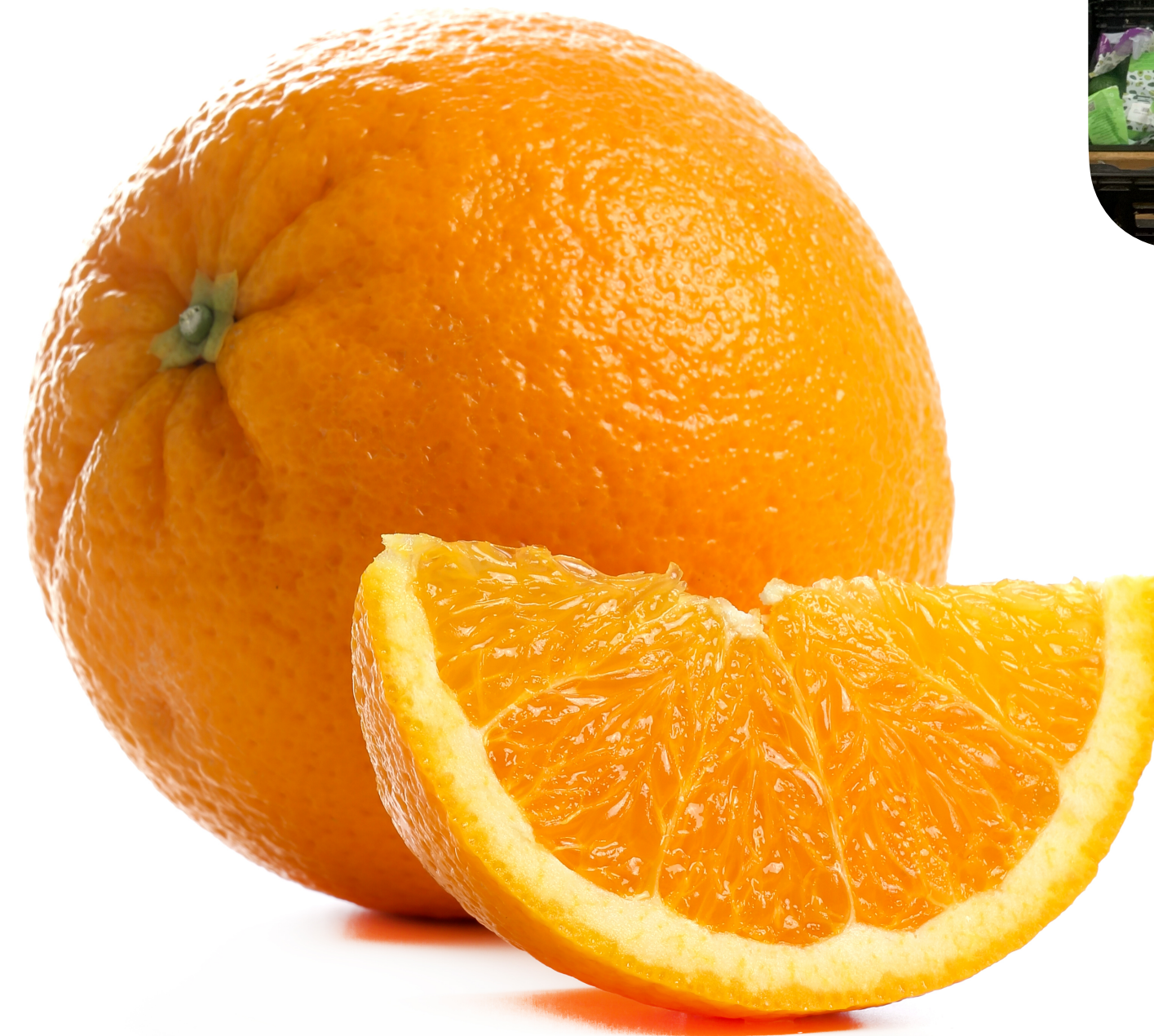
x_1

- VS -

Oranges



$$c_2 = 1.49$$



x_2

Consider a utility maximization problem with the Cobb-Douglas utility function, $U(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$, subject to a budget constraint. The Lagrangian is:

$$\mathcal{L}(x, \lambda) = x_1^\alpha x_2^{1-\alpha} + \lambda \cdot (I - c_1 x_1 - c_2 x_2)$$

which gives the first-order optimality conditions:

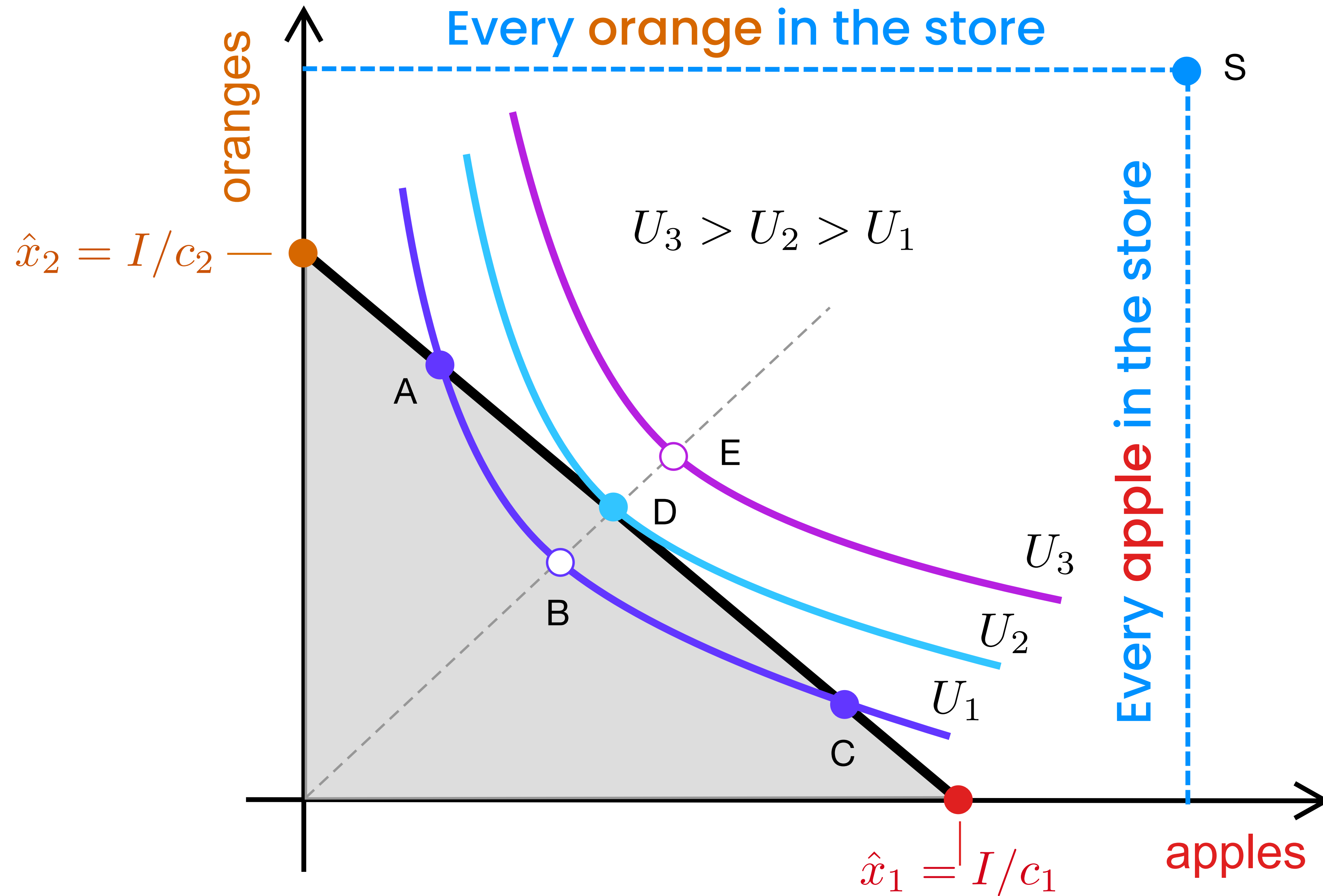
$$\frac{\partial \mathcal{L}}{\partial x_1} = \alpha \cdot x_1^{\alpha-1} x_2^{1-\alpha} - \lambda \cdot c_1 = 0$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = (1 - \alpha) \cdot x_1^\alpha x_2^{-\alpha} - \lambda \cdot c_2 = 0$$

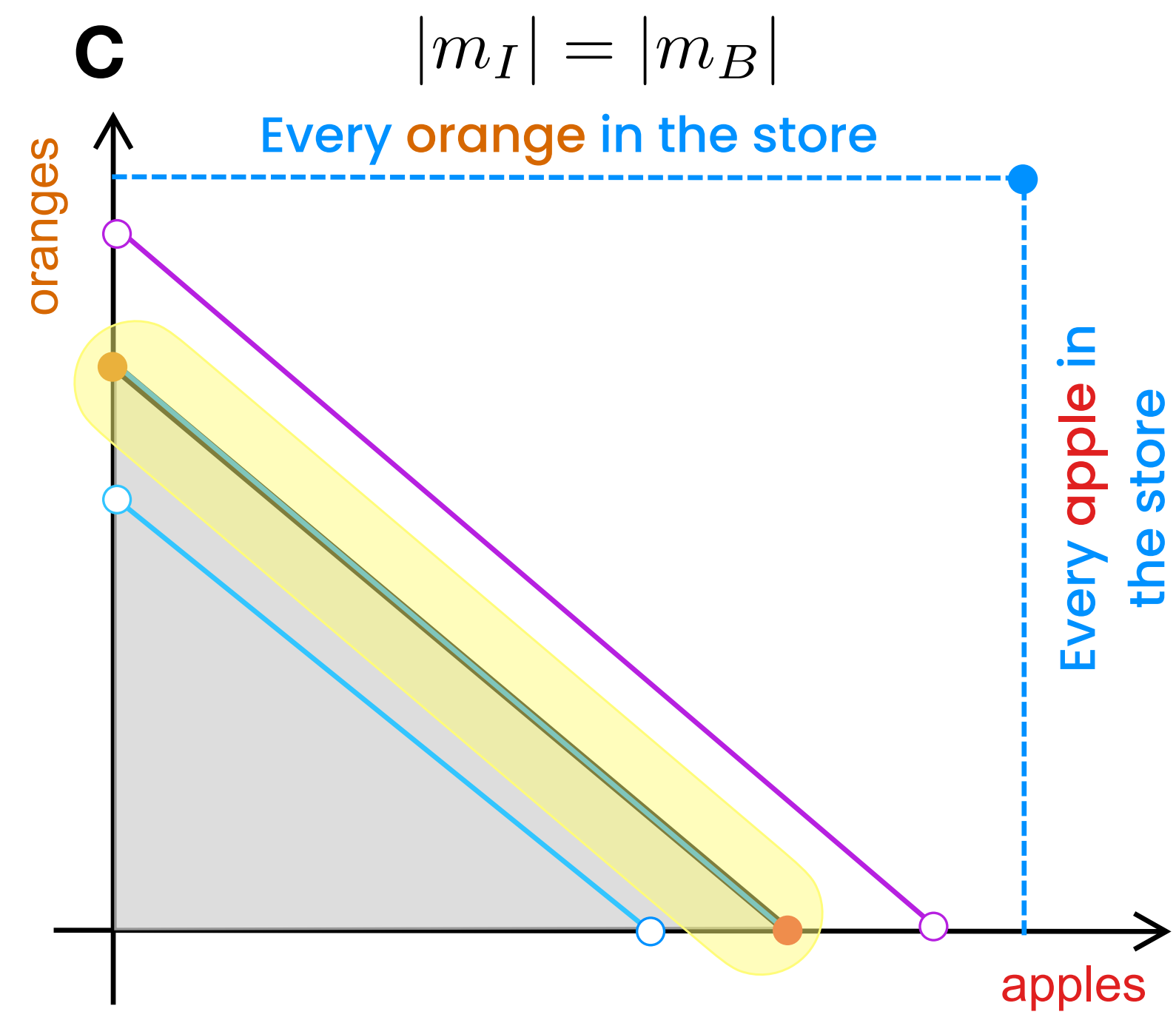
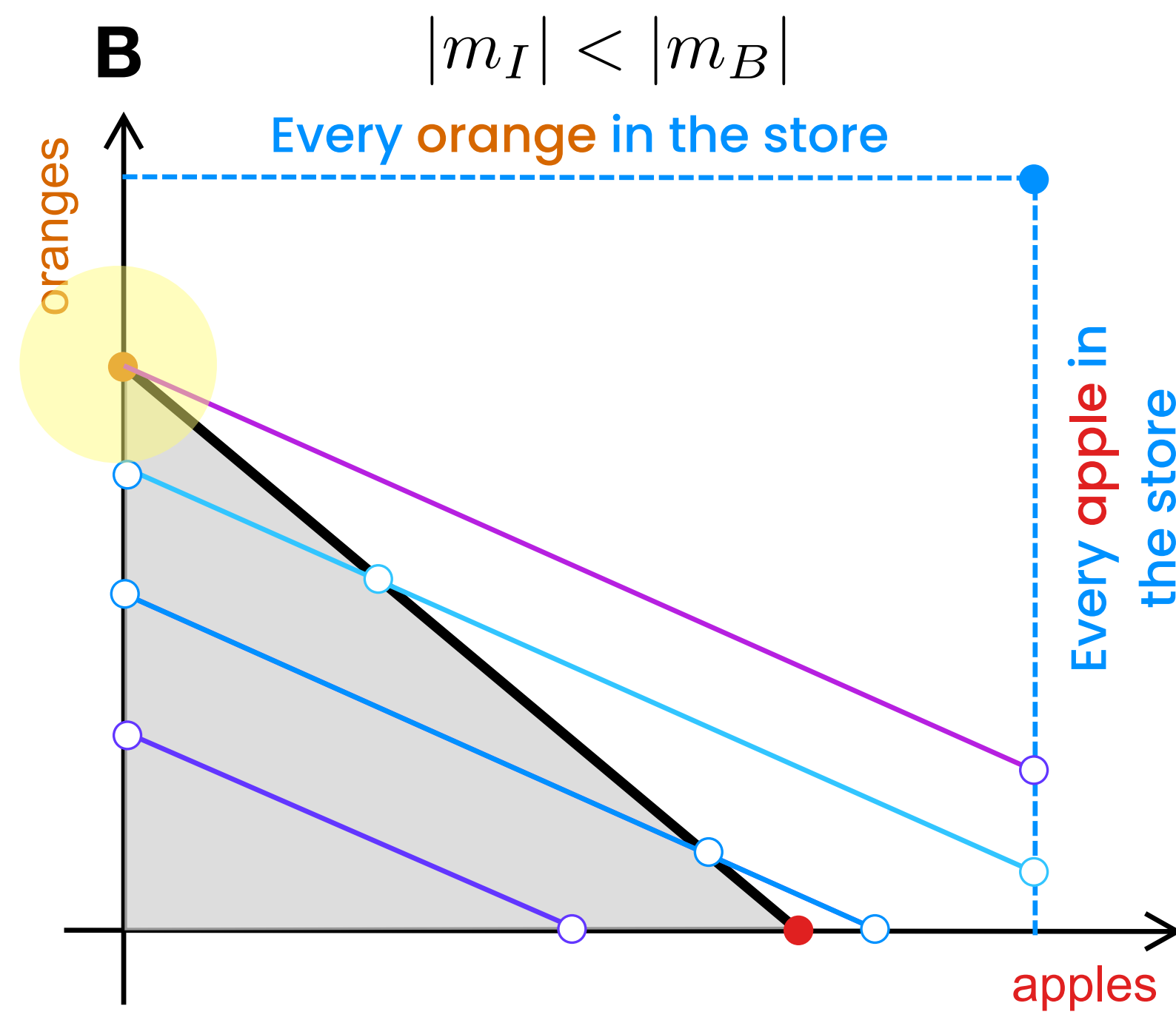
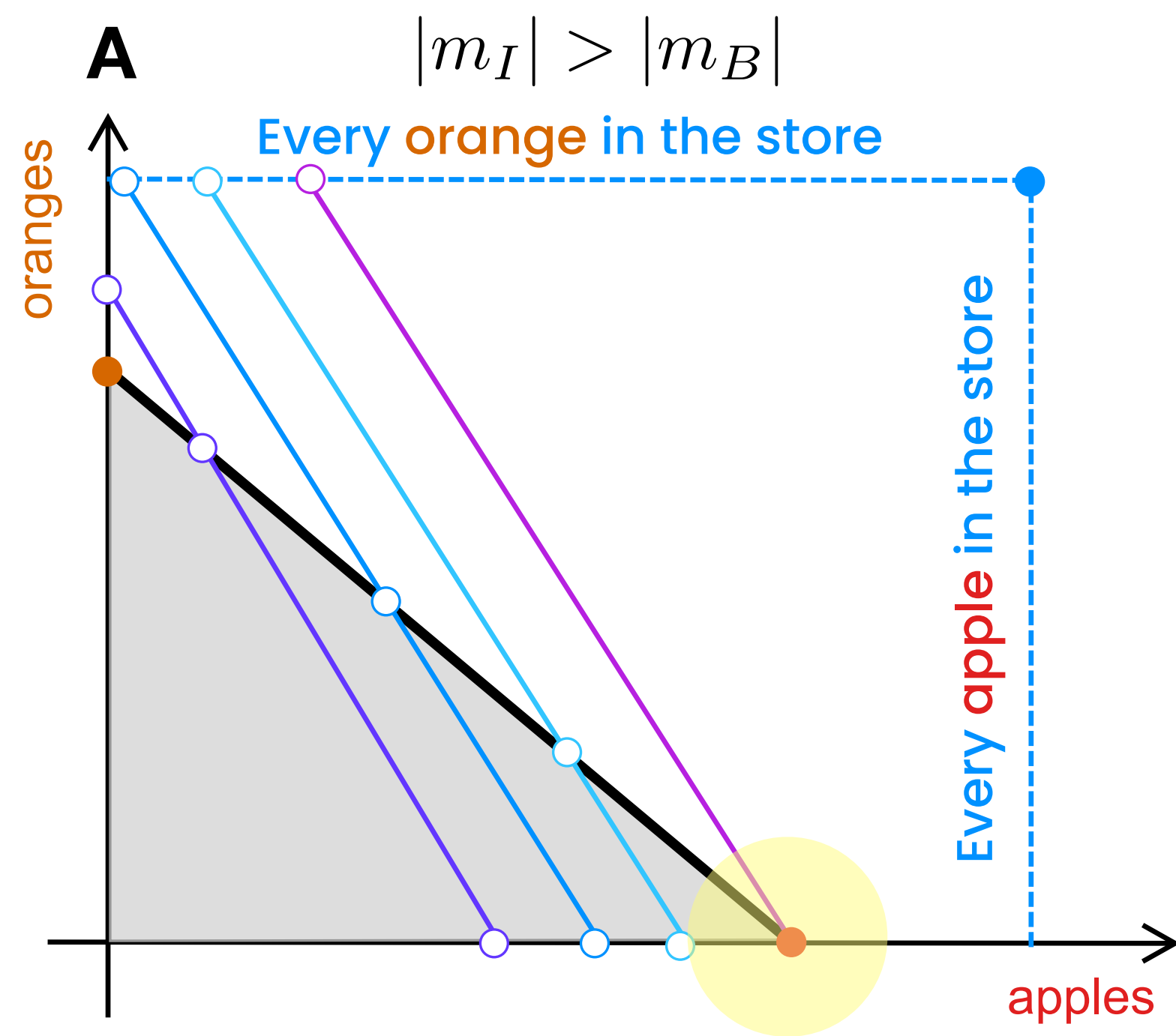
$$\frac{\partial \mathcal{L}}{\partial \lambda} = I - c_1 x_1 - c_2 x_2 = 0$$

This system can be solved for the optimal consumption levels of x_1 and x_2 . However, it is non-linear. Thus, we often need to use numerical methods to find the optimal values of x_1 and x_2 .

Utility: Cobb-Douglas

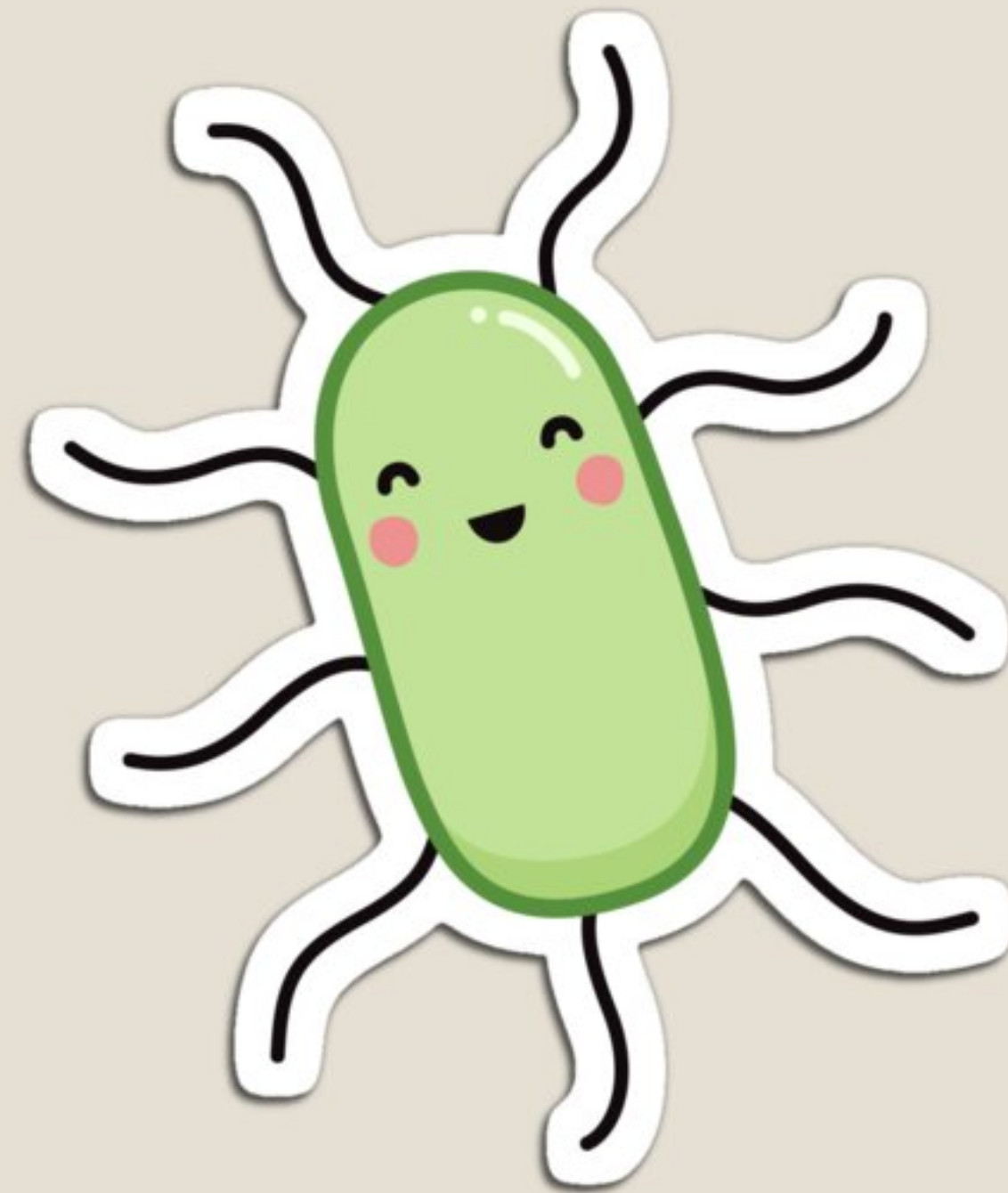


Example: CHEME-5760-L3b-OptimalSoln-CobbDouglas-Utility.ipynb

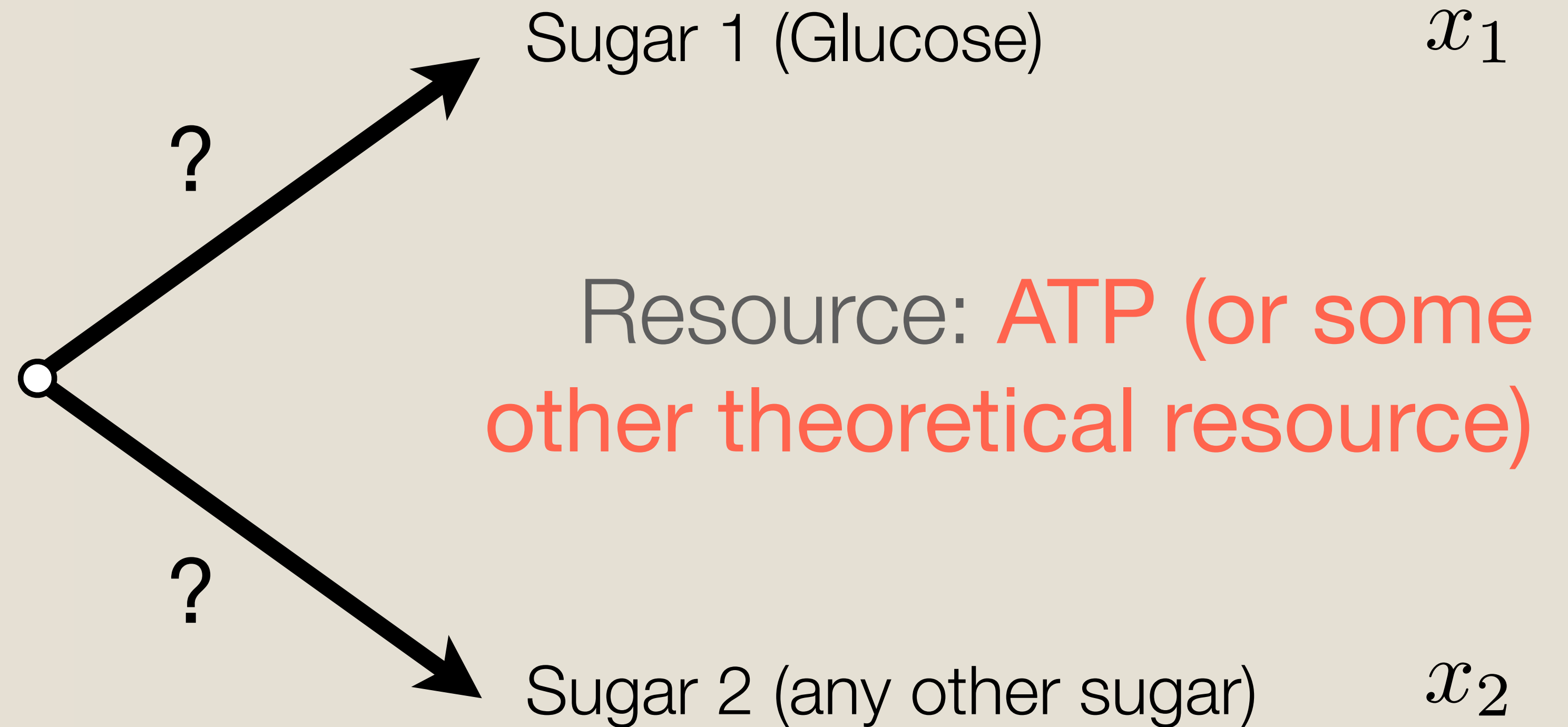


Example: CHEME-5760-OptimalSoln-Linear-Utility.ipynb

Utility: Does *E. coli* have free will?



Escherichia coli



Kompala DS, Ramkrishna D, Jansen NB, Tsao GT. (1986) Investigation of bacterial growth on mixed substrates: experimental evaluation of cybernetic models. *Biotechnol Bioeng.* 28:1044-55. doi: 10.1002/bit.260280715. [PMID: 18555426](#).

Example: CHEME-5760-L3b-OptimalSoln-KompalaModel.ipynb

Today?